

Assignment #1 - Fourier Series

Due Tuesday, September 8, 2026

1. (15 points) Let f be a 2π -periodic function satisfying $f(x) = x$ for $x \in (-\pi, \pi)$.
(a) Compute the Fourier series of f and show all work.

Solution — The Fourier series of a 2π -periodic function on $[-\pi, \pi]$ is given by

$$f(x) = \frac{1}{2}a_0 + \sum_{n=1}^{\infty} a_n \cos(nx) + \sum_{n=1}^{\infty} b_n \sin(nx)$$

where

$$a_0 = \frac{1}{\pi} \int_a^b f(x) dx \quad a_n = \frac{1}{\pi} \int_a^b f(x) \cos(nx) dx$$

$$b_n = \frac{1}{\pi} \int_a^b f(x) \sin(nx) dx$$

Then for our specific 2π -periodic function $f(x) = x$, we have

$$a_0 = \frac{1}{\pi} \int_{-\pi}^{\pi} x dx = 0 \quad (\text{by oddness})$$

$$a_n = \frac{1}{\pi} \int_{-\pi}^{\pi} x \cos(nx) dx = 0 \quad (\text{also by oddness})$$

$$\begin{aligned} b_n &= \frac{1}{\pi} \int_{-\pi}^{\pi} x \sin(nx) dx \\ &= \frac{2}{\pi} \int_0^{\pi} x \sin(nx) dx \quad (\text{by evenness}) \\ &= \frac{2}{\pi} \left[\frac{-x}{n} \cos(nx) \Big|_0^{\pi} + \frac{1}{n} \int_0^{\pi} \cos(nx) dx \right] \quad (\text{by IBP}) \\ &= \frac{2}{\pi} \left[\frac{-\pi}{n} \cos(n\pi) + \frac{\sin(n\pi)}{n^2} \right] \\ &= \frac{2 \sin(n\pi)}{\pi n^2} - \frac{2}{n} \cos(n\pi) \end{aligned}$$

Thus our series expansion is

$$f(x) = \sum_{n=1}^{\infty} \left[\frac{2 \sin(n\pi)}{\pi n^2} - \frac{2}{n} \cos(nx) \right] \sin(nx)$$

Note that $\sin(n\pi) = 0$ and $\cos(n\pi) = (-1)^n$ for $n = 1, 2, 3, \dots$. Hence, we have

$$f(x) = 2 \sum_{n=1}^{\infty} \frac{(-1)^{n+1} \sin(nx)}{n}$$

- (b) Using your answer from (a), graph by hand on the same axes the three functions $f(x)$, $g(x)$, and $h(x)$ on the interval $[-\pi, \pi]$ where

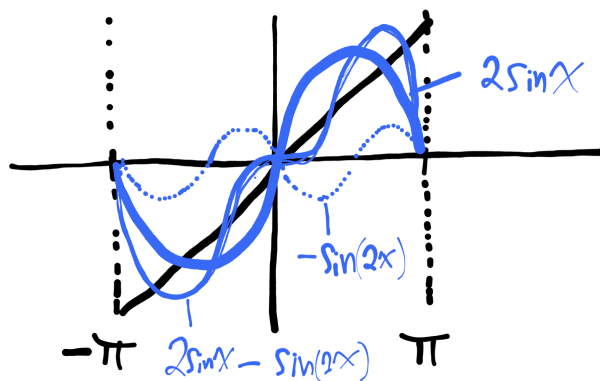
$$g(x) = b_1 \sin(x)$$

and

$$h(x) = b_1 \sin(x) + b_2 \sin(2x)$$

are the one-term and two-term Fourier Series approximations of f , respectively.

Solution — We have $g(x) = 2 \sin(x)$ and $h(x) = 2 \sin(x) - \sin(2x)$



2. (5 points) Let $f(x)$ be a given function for $x \in [-\pi, \pi]$ and find $c \in \mathbb{R}$ which minimizes

$$\int_{-\pi}^{\pi} (f(x) - c)^2 dx.$$

Note that your answer should depend in some manner upon the given function f .

Solution — Let us find the c that minimizes

$$D(c) = \int_{-\pi}^{\pi} (f(x) - c)^2$$

Note that $D(c)$ appears to be some of sort of distance-squared metric measuring the accumulated distance between $f(x)$ and the line $y = c$. So intuitively, we expect c has to be something like the average value of $f(x)$ in order for all of f to be as close as possible to c —that is, to minimize this distance. We will use the first derivative test.

$$\frac{dD}{dc} = \frac{d}{dc} \int_{-\pi}^{\pi} (f(x) - c)^2 dx$$

We can bring the derivative inside the integral because it is independent of the variable of integration x . Doing this will translate the derivative into a partial derivative, because the integrand is a function of both x and the parameter c .

$$\begin{aligned} &= \int_{-\pi}^{\pi} \frac{\partial}{\partial c} (f(x) - c)^2 dx \\ &= \int_{-\pi}^{\pi} 2(f(x) - c)(0 - 1) dx \\ &= 2 \int_{-\pi}^{\pi} (c - f(x)) dx \\ &= 4\pi c - 2 \int_{-\pi}^{\pi} f(x) dx = 0 \\ &\implies \boxed{c = \frac{1}{2\pi} \int_{-\pi}^{\pi} f(x) dx} \end{aligned}$$

So the value of c that minimizes $D(c)$ happens to be exactly the average value of $f(x)$ over $[-\pi, \pi]$, as predicted. We verify with the second-derivative test:

$$\begin{aligned} &= \frac{d^2 D}{dc^2} = \frac{d}{dc} \left(2 \int_{-\pi}^{\pi} (c - f(x)) dx \right) \\ &= 4\pi > 0 \quad \text{for all } c \end{aligned}$$

3. (10 points) Let $f(x)$ be a given function for $x \in [-\pi, \pi]$ and find $a, b, c \in \mathbb{R}$ which minimize

$$\int_{-\pi}^{\pi} (f(x) - a \cos(x) - b \sin(x) - c)^2 dx.$$

Again, note that your answer should depend in some manner upon the given function f and use the analogous hint as above for a multivariate function.

Solution — Let

$$F(a, b, c) = \int_{-\pi}^{\pi} (f(x) - a \cos(x) - b \sin(x) - c)^2 dx$$

Then we would like to minimize F . We find where the gradient of F is zero:

$$\nabla F(a, b, c) = \left\langle \frac{\partial F}{\partial a}, \frac{\partial F}{\partial b}, \frac{\partial F}{\partial c} \right\rangle = \mathbf{0}$$

$$\begin{aligned} \frac{\partial F}{\partial a} &= \frac{\partial}{\partial a} \int_{-\pi}^{\pi} (f(x) - a \cos(x) - b \sin(x) - c)^2 dx \\ &= \int_{-\pi}^{\pi} 2(f(x) - a \cos(x) - b \sin(x) - c)(-\cos(x)) dx \\ &= 2 \left[- \int_{-\pi}^{\pi} f(x) \cos(x) dx + a \int_{-\pi}^{\pi} \cos^2(x) dx + b \int_{-\pi}^{\pi} \sin(x) \cos(x) dx + c \int_{-\pi}^{\pi} \cos(x) dx \right] \\ &= 2 \left[- \int_{-\pi}^{\pi} f(x) \cos(x) dx + \pi a + 0 + 0 \right] = 0 \\ \implies a &= \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \cos(x) dx \end{aligned}$$

$$\begin{aligned} \frac{\partial F}{\partial b} &= \frac{\partial}{\partial b} \int_{-\pi}^{\pi} (f(x) - a \cos(x) - b \sin(x) - c)^2 dx \\ &= \int_{-\pi}^{\pi} 2(f(x) - a \cos(x) - b \sin(x) - c)(-\sin(x)) dx \\ &= 2 \left[- \int_{-\pi}^{\pi} f(x) \sin(x) dx + a \int_{-\pi}^{\pi} \sin(x) \cos(x) dx + b \int_{-\pi}^{\pi} \sin^2(x) dx + c \int_{-\pi}^{\pi} \sin(x) dx \right] \\ &= 2 \left[- \int_{-\pi}^{\pi} f(x) \sin(x) dx + 0 + \pi b + 0 \right] = 0 \\ \implies b &= \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \sin(x) dx \end{aligned}$$

$$\begin{aligned} \frac{\partial F}{\partial c} &= \frac{\partial}{\partial c} \int_{-\pi}^{\pi} (f(x) - a \cos(x) - b \sin(x) - c)^2 dx \\ &= \int_{-\pi}^{\pi} 2(f(x) - a \cos(x) - b \sin(x) - c)(-1) dx \\ &= 2 \left[- \int_{-\pi}^{\pi} f(x) dx + a \int_{-\pi}^{\pi} \cos(x) dx + b \int_{-\pi}^{\pi} \sin(x) dx + c \int_{-\pi}^{\pi} dx \right] \\ &= 2 \left[- \int_{-\pi}^{\pi} f(x) dx + 0 + 0 + 2\pi c \right] = 0 \\ \implies c &= \frac{1}{2\pi} \int_{-\pi}^{\pi} f(x) \sin(x) dx \end{aligned}$$

A triplet that minimizes F is thus $(a, b, c) = (a_1, b_1, f_{avg})$, where a_1 is the first cosine Fourier coefficient, b_1 is the first sine Fourier coefficient, and f_{avg} is the average value of $f(x)$ over $[-\pi, \pi]$.

We can make a rudimentary argument for why such (a, b, c) is the global minimum: in order for the squared difference error between $f(x)$ and $g(x) = a \cos(x) + b \sin(x) + c$ to be minimized, $g(x)$ has to be an accurate approximation for $f(x)$, which is attained, for instance, from the first-order Fourier approximation of f . Changing $g(x)$ in any way results in increasing error, where $g(x)$ becomes a poorer approximation of f , causing squared error to increase in all such directions. Hence we have found the minimum such (a, b, c) .